

AWS CLI COMMAND CHEAT SHEET

Essential Commands for Cloud & DevOps Engineers

Pro Tip: Replace placeholder values (xxxx, account-id, region, etc.) with your actual AWS resource IDs and values.

General Configuration

► *Check AWS CLI version*

aws --version

► *Configure AWS credentials and settings*

aws configure

► *List configuration data*

aws configure list

► *Verify your identity*

aws sts get-caller-identity

► *Get help for any command*

aws help

aws s3 help

IAM (Identity and Access Management)

User Management

► *List all IAM users*

aws iam list-users

► *Get current user details*

aws iam get-user

► *Create a new IAM user*

aws iam create-user --user-name newuser

Permissions & Policies

► *List all IAM policies*

aws iam list-policies

► *Attach policy to user*

aws iam attach-user-policy --user-name username --policy-arn arn:aws:iam::aws:policy/ReadOnlyAccess

► *List access keys for a user*

aws iam list-access-keys --user-name username

Groups & Profiles

► *List IAM groups*

```
aws iam list-groups --query
```

► *List instance profiles*

```
aws iam list-instance-profiles
```

S3 (Simple Storage Service)

Bucket Operations

► *List all buckets*

```
aws s3 ls
```

```
aws s3api list-buckets
```

► *Create a new bucket*

```
aws s3 mb s3://my-bucket-name
```

► *Delete a bucket (with force to remove contents)*

```
aws s3 rb s3://my-bucket-name --force
```

► *List contents of a bucket*

```
aws s3 ls s3://my-bucket-name
```

File Operations

► *Copy file to S3*

```
aws s3 cp my-file.txt s3://my-bucket-name/
```

► *Copy file from S3*

```
aws s3 cp s3://my-bucket-name/my-file.txt .
```

► *Remove a specific file*

```
aws s3 rm s3://my-bucket-name/file.txt
```

► *Sync local directory to S3*

```
aws s3 sync ./local-folder s3://my-bucket-name/folder
```

Advanced Features

► *Set bucket versioning*

```
aws s3api put-bucket-versioning --bucket my-bucket --versioning-configuration Status=Enabled
```

► *Get bucket policy*

```
aws s3api get-bucket-policy --bucket my-bucket-name
```

EC2 (Elastic Compute Cloud)

Instance Management

▶ *List all EC2 instances*

```
aws ec2 describe-instances
```

▶ *List running instances only*

```
aws ec2 describe-instances --filters "Name=instance-state-name,Values=running"
```

▶ *Start an instance*

```
aws ec2 start-instances --instance-ids i-1234567890abcdef0
```

▶ *Stop an instance*

```
aws ec2 stop-instances --instance-ids i-1234567890abcdef0
```

▶ *Terminate an instance*

```
aws ec2 terminate-instances --instance-ids i-1234567890abcdef0
```

Key Pairs & Launch

▶ *Create a key pair*

```
aws ec2 create-key-pair --key-name mykey > mykey.pem
```

▶ *Launch a new instance*

```
aws ec2 run-instances \  
--image-id ami-0f58b397bc5c1f2e8 \  
--instance-type t2.micro \  
--key-name mykey \  
--security-group-ids sg-xxxx \  
--subnet-id subnet-xxxx \  
--count 1
```

▶ *Describe available AMIs*

```
aws ec2 describe-images --owners self amazon
```

Security Groups

▶ *List security groups*

```
aws ec2 describe-security-groups
```

▶ *Create security group*

```
aws ec2 create-security-group --group-name my-sg --description "My security group" --vpc-id vpc-xxxx
```

▶ *Add inbound rule (SSH access)*

```
aws ec2 authorize-security-group-ingress --group-id sg-xxxx --protocol tcp --port 22 --cidr 0.0.0.0/0
```

Storage & Snapshots

▶ *List EBS volumes*

```
aws ec2 describe-volumes
```

▶ *Create snapshot*

```
aws ec2 create-snapshot --volume-id vol-xxxx --description "My snapshot"
```

► *List snapshots*

aws ec2 describe-snapshots --owner-ids self

Elastic IP

► *List elastic IPs*

aws ec2 describe-addresses

► *Allocate elastic IP*

aws ec2 allocate-address --domain vpc

► *Get instance metadata (run from EC2 instance)*

curl http://169.254.169.254/latest/meta-data/



VPC (Virtual Private Cloud)

VPC & Subnets

► *List all VPCs*

aws ec2 describe-vpcs

► *Create a VPC*

aws ec2 create-vpc --cidr-block 10.0.0.0/16

► *List subnets*

aws ec2 describe-subnets

► *Create a subnet*

aws ec2 create-subnet --vpc-id vpc-xxxx --cidr-block 10.0.1.0/24

Routing & Gateways

► *List route tables*

aws ec2 describe-route-tables

► *Create internet gateway*

aws ec2 create-internet-gateway

► *Attach internet gateway to VPC*

aws ec2 attach-internet-gateway --internet-gateway-id igw-xxxx --vpc-id vpc-xxxx

► *List internet gateways*

aws ec2 describe-internet-gateways

NAT & Endpoints

► *List NAT gateways*

aws ec2 describe-nat-gateways

► *List VPC endpoints*

aws ec2 describe-vpc-endpoints

Auto Scaling

► *List Auto Scaling groups*

aws autoscaling describe-auto-scaling-groups

► *Describe Launch templates*

aws ec2 describe-launch-templates

► *Create Launch template*

aws ec2 create-launch-template --launch-template-name my-template --version-description "v1" --launch-template-data file://template.json

► *Update Auto Scaling group capacity*

aws autoscaling set-desired-capacity --auto-scaling-group-name my-asg --desired-capacity 3

ELB (Elastic Load Balancing)

► *List Application/Network Load Balancers*

aws elbv2 describe-load-balancers

► *List target groups*

aws elbv2 describe-target-groups

► *Describe target health*

aws elbv2 describe-target-health --target-group-arn arn:aws:elasticloadbalancing:region:account-id:targetgroup/my-targets/xxxx

► *Register targets*

aws elbv2 register-targets --target-group-arn arn --targets Id=i-xxxx

CloudWatch

Metrics & Statistics

► *List metrics*

aws cloudwatch list-metrics

► *Get metric statistics*

aws cloudwatch get-metric-statistics \
--metric-name CPUUtilization \
--namespace AWS/EC2 \
--statistics Average \
--period 300 \
--start-time 2024-11-25T00:00:00Z \
--end-time 2024-11-25T23:59:59Z

Alarms

► *Create metric alarm*

aws cloudwatch put-metric-alarm \

```
--alarm-name "High-CPU-Alarm" \  
--metric-name CPUUtilization \  
--namespace AWS/EC2 \  
--statistic Average \  
--period 300 \  
--threshold 80 \  
--comparison-operator GreaterThanThreshold \  
--evaluation-periods 2
```

► *List alarms*

```
aws cloudwatch describe-alarms
```

► *Delete alarm*

```
aws cloudwatch delete-alarms --alarm-names "alarm-name"
```

CloudWatch Logs

► *List Log groups*

```
aws logs describe-log-groups
```

► *List Log streams*

```
aws logs describe-log-streams --log-group-name /aws/lambda/my-function
```

► *Get Log events*

```
aws logs get-log-events --log-group-name /aws/lambda/my-function --log-stream-name  
2024/11/25/[$LATEST]xxxxx
```

► *Create Log group*

```
aws logs create-log-group --log-group-name my-log-group
```

► *Tail Logs (follow in real-time)*

```
aws logs tail /aws/lambda/my-function --follow
```

SNS (Simple Notification Service)

► *List topics*

```
aws sns list-topics
```

► *Create a topic*

```
aws sns create-topic --name my-alert-topic
```

► *Subscribe to a topic*

```
aws sns subscribe \  
--topic-arn arn:aws:sns:us-east-1:123456789012:my-alert-topic \  
--protocol email \  
--notification-endpoint myemail@example.com
```

► *Publish message to topic*

```
aws sns publish --topic-arn arn:aws:sns:us-east-1:123456789012:my-topic --message "Alert message"
```

► *List subscriptions*

aws sns list-subscriptions

 Lambda

► *List Lambda functions*

aws lambda list-functions

► *Get function configuration*

aws lambda get-function --function-name myfunction

► *Invoke a function*

aws lambda invoke --function-name myfunction response.json

► *Create function*

aws lambda create-function --function-name myfunction --runtime python3.9 --role
arn:aws:iam::account:role/lambda-role --handler index.handler --zip-file fileb://function.zip

► *Update function code*

aws lambda update-function-code --function-name myfunction --zip-file fileb://function.zip

► *List function versions*

aws lambda list-versions-by-function --function-name myfunction

 CloudFormation

► *List stacks*

aws cloudformation list-stacks

► *Deploy/create stack*

aws cloudformation deploy \
--template-file my-stack.yaml \
--stack-name mystack

► *Describe stack*

aws cloudformation describe-stacks --stack-name mystack

► *Validate template*

aws cloudformation validate-template --template-body file://template.yaml

► *Get stack events*

aws cloudformation describe-stack-events --stack-name mystack

► *Delete stack*

aws cloudformation delete-stack --stack-name mystack

 Route 53

► *List hosted zones*

aws route53 list-hosted-zones

► *List resource record sets*

aws route53 list-resource-record-sets --hosted-zone-id Z1234567890ABC

► *Create hosted zone*

aws route53 create-hosted-zone --name example.com --caller-reference 2024-11-25-001

► *Change resource record sets*

aws route53 change-resource-record-sets --hosted-zone-id Z1234567890ABC --change-batch file://changes.json

 CloudFront

► *List distributions*

aws cloudfront list-distributions

► *Get distribution details*

aws cloudfront get-distribution --id E1234567890ABC

► *Create invalidation (cache purge)*

aws cloudfront create-invalidation --distribution-id E1234567890ABC --paths "/*"

► *List invalidations*

aws cloudfront list-invalidations --distribution-id E1234567890ABC

 KMS (Key Management Service)

► *List keys in region*

aws kms list-keys --region us-east-1

► *Describe key*

aws kms describe-key --key-id alias/my-key

► *Create key*

aws kms create-key --description "My encryption key"

► *Create alias*

aws kms create-alias --alias-name alias/my-key --target-key-id key-id

► *Encrypt data*

aws kms encrypt --key-id alias/my-key --plaintext fileb://secret.txt --output text --query CiphertextBlob

► *Decrypt data*

aws kms decrypt --ciphertext-blob fileb://encrypted.txt --output text --query Plaintext

 WAF (Web Application Firewall)

► *List Web ACLs*

aws wafv2 list-web-acls --scope REGIONAL --region us-east-1

► *Get Web ACL details*

aws wafv2 get-web-acl --name my-web-acl --scope REGIONAL --id xxxx --region us-east-1

► *List IP sets*

aws wafv2 list-ip-sets --scope REGIONAL --region us-east-1



RDS (Relational Database Service)

Instance Management

► *List DB instances*

aws rds describe-db-instances

► *Start DB instance*

aws rds start-db-instance --db-instance-identifier mydb

► *Stop DB instance*

aws rds stop-db-instance --db-instance-identifier mydb

► *Modify DB instance*

aws rds modify-db-instance --db-instance-identifier mydb --allocated-storage 100 --apply-immediately

Snapshots

► *Create DB snapshot*

aws rds create-db-snapshot --db-instance-identifier mydb --db-snapshot-identifier mydb-snapshot

► *List DB snapshots*

aws rds describe-db-snapshots



DynamoDB

► *List tables*

aws dynamodb list-tables

► *Describe table*

aws dynamodb describe-table --table-name my-table

► *Get item*

aws dynamodb get-item --table-name my-table --key '{"id":{"S":"123"}}'

► *Put item*

aws dynamodb put-item --table-name my-table --item '{"id":{"S":"123"},"name":{"S":"John"}}'

► *Scan table*

aws dynamodb scan --table-name my-table

► *Query table*

```
aws dynamodb query --table-name my-table --key-condition-expression "id = :id" --expression-attribute-values '{"id":{"S":"123"}}'
```

ECS (Elastic Container Service)

► *List clusters*

```
aws ecs list-clusters
```

► *List services*

```
aws ecs list-services --cluster my-cluster
```

► *List tasks*

```
aws ecs list-tasks --cluster my-cluster
```

► *Describe service*

```
aws ecs describe-services --cluster my-cluster --services my-service
```

► *Update service*

```
aws ecs update-service --cluster my-cluster --service my-service --desired-count 3
```

ECR (Elastic Container Registry)

► *List repositories*

```
aws ecr describe-repositories
```

► *Create repository*

```
aws ecr create-repository --repository-name my-app
```

► *Get login password for Docker*

```
aws ecr get-login-password --region us-east-1 | docker login --username AWS --password-stdin account-id.dkr.ecr.us-east-1.amazonaws.com
```

► *List images*

```
aws ecr list-images --repository-name my-app
```

Systems Manager (SSM)

Parameters

► *List parameters*

```
aws ssm describe-parameters
```

► *Get parameter value*

```
aws ssm get-parameter --name /my-app/config --with-decryption
```

► *Put parameter*

```
aws ssm put-parameter --name /my-app/config --value "config-value" --type SecureString
```

Session Manager

► *Start interactive session*

```
aws ssm start-session --target i-1234567890abcdef0
```

► *Send command to instances*

```
aws ssm send-command --document-name "AWS-RunShellScript" --targets "Key=instanceids,Values=i-xxxx" --parameters 'commands=["echo hello"]'
```



Cost Explorer & Billing

► *Get cost and usage*

```
aws ce get-cost-and-usage --time-period Start=2024-11-01,End=2024-11-30 --granularity MONTHLY --metrics BlendedCost
```

► *Get cost forecast*

```
aws ce get-cost-forecast --time-period Start=2024-12-01,End=2024-12-31 --metric BLENDED_COST --granularity MONTHLY
```



Useful Tips & Tricks

Output Formatting

► *JSON output (default)*

```
aws ec2 describe-instances
```

► *Table format (human-readable)*

```
aws ec2 describe-instances --output table
```

► *Text format*

```
aws ec2 describe-instances --output text
```

► *YAML format*

```
aws ec2 describe-instances --output yaml
```

Query & Filter with JMESPath

► *Use --query with JMESPath for specific data*

```
aws ec2 describe-instances --query 'Reservations[*].Instances[*].[InstanceId,State.Name]'
```

► *Use --filters to narrow results*

```
aws ec2 describe-instances --filters "Name=instance-type,Values=t2.micro"
```

► *Combine query and filters*

```
aws ec2 describe-instances --filters "Name=instance-state-name,Values=running" --query 'Reservations[*].Instances[*].[InstanceId,PublicIpAddress]' --output table
```

Profile & Region Management

► *Use specific AWS profile*

```
aws s3 ls --profile production
```

► *Configure new profile*

```
aws configure --profile development
```

► *Set default region*

aws configure set region us-west-2

Dry Run Mode

► *Test command without executing (where supported)*

aws ec2 run-instances --dry-run --image-id ami-xxxx --instance-type t2.micro

 Quick Reference

Service	Code	Icon
Identity & Access Management	IAM	
Simple Storage Service	S3	
Elastic Compute Cloud	EC2	
Virtual Private Cloud	VPC	
CloudWatch	CW	
Lambda	λ	
RDS	DB	
DynamoDB	NoSQL	
ECS/ECR	Containers	
SNS	Notifications	
CloudFormation	IaC	
Route 53	DNS	
CloudFront	CDN	
KMS	Encryption	

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AWS Region: us-east-1

 [**AWS Official Documentation**](https://docs.aws.amazon.com/)
